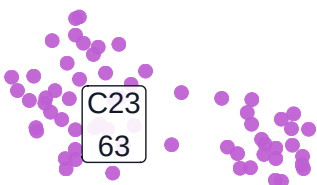
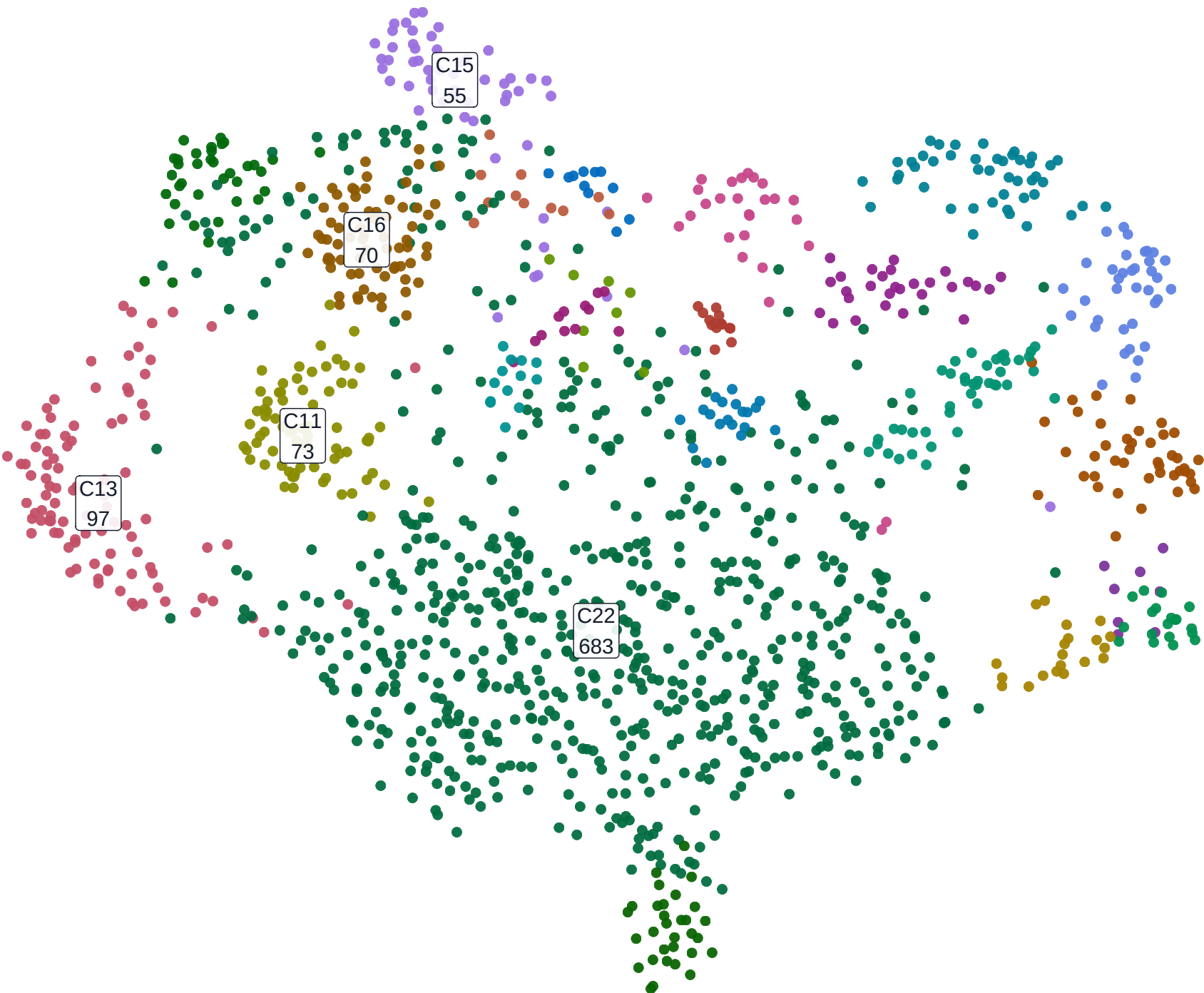


Goncalves TBS clustering result on the adaptive-diffusion tree

Same cells on UMAP and same full tree; this is the clustering result used for the progenitor analysis below

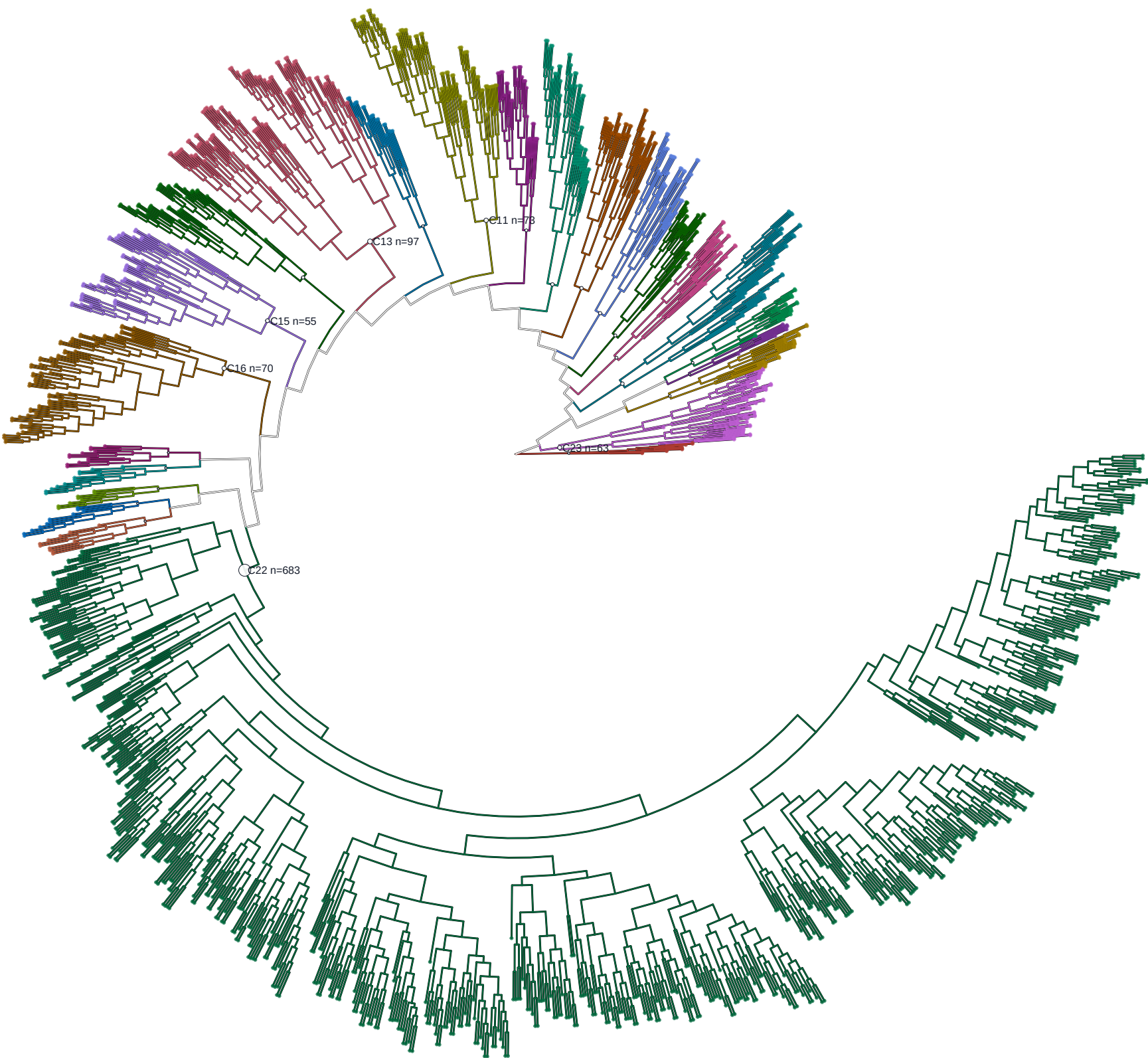
UMAP by TBS cluster

24 clusters from adaptive-diffusion topology TBS; labels shown for clusters >= 50 cells



Goncalves adaptive-diffusion TBS tree by final cluster

Colored branches are exact final cluster subtrees; pale grey marks shared ancestors



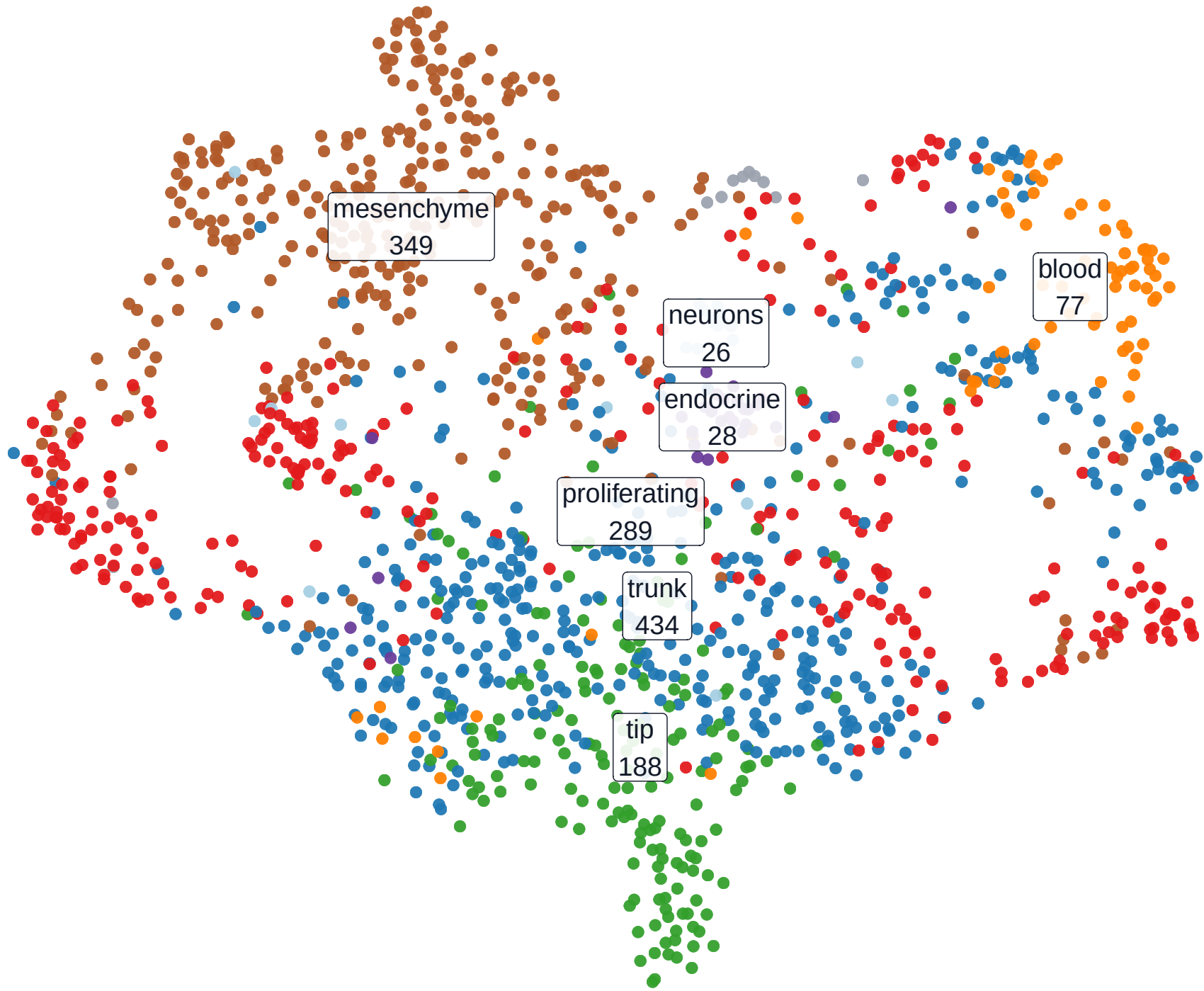
Cluster size ○ 200 ○ 400 ○ 600

Goncalves fetal population UMAP and full TBS tree

Cell UMAP and matching whole radial adaptive-diffusion TBS tree

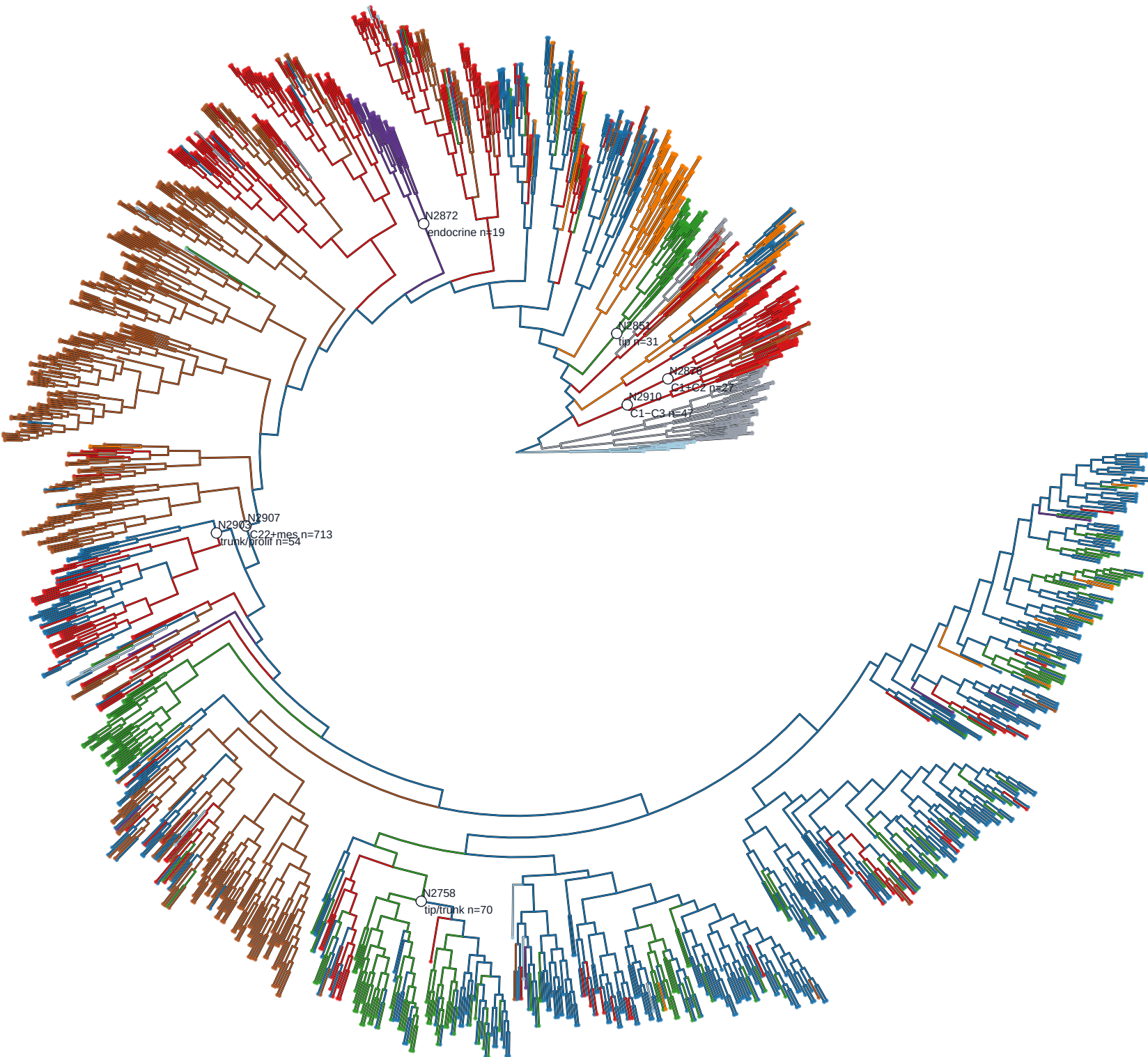
UMAP by fetal population

Same cell labels used for terminal tree tips



Goncalves adaptive-diffusion TBS tree by dominant fetal population

Whole tree; labels mark progenitor/endocrine nodes from the signature analysis



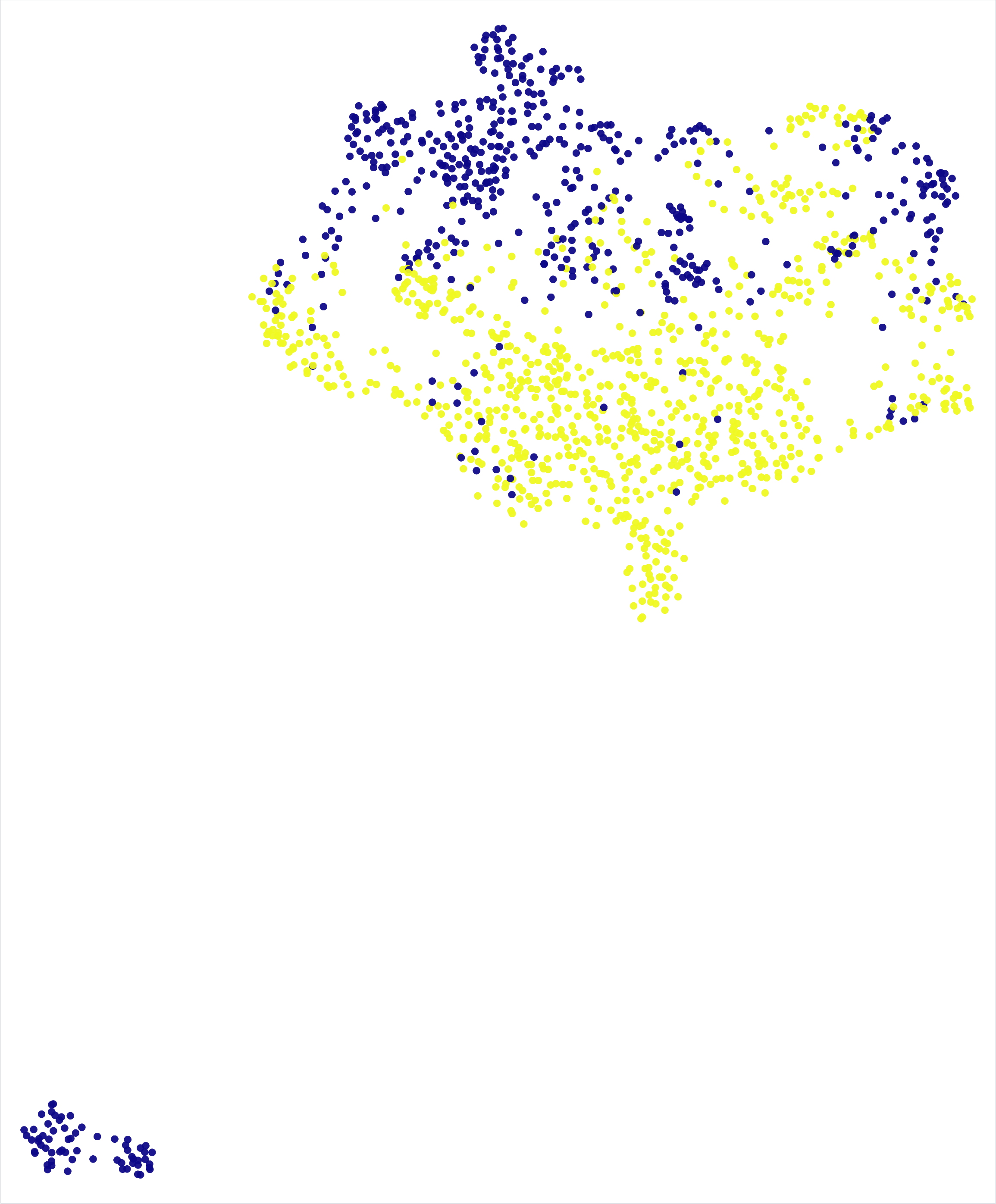
trunk proliferating mesenchyme neurons
tip endocrine blood unknown

Goncalves progenitor-fraction UMAP and full TBS tree

Cell-level trunk/tip/proliferating membership beside descendant fraction on the whole tree

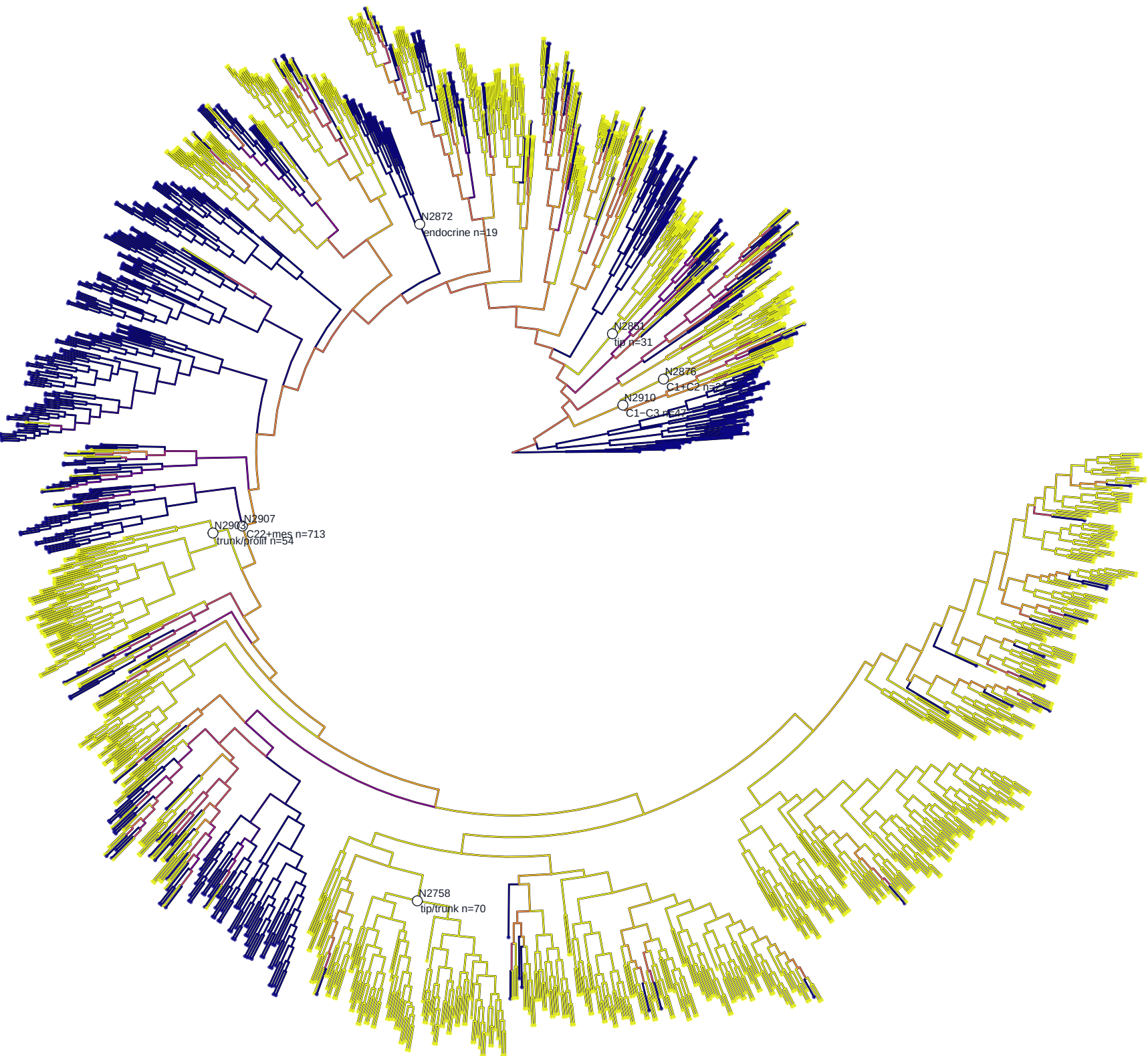
UMAP by progenitor-label membership

Tree color is descendant fraction; cell-level UMAP is 0/1 membership



Goncalves adaptive-diffusion TBS tree by progenitor fraction

Continuous color is fraction of descendant cells labeled trunk, tip, or proliferating



Cell is trunk/tip/prolif
0.00 0.25 0.50 0.75 1.00

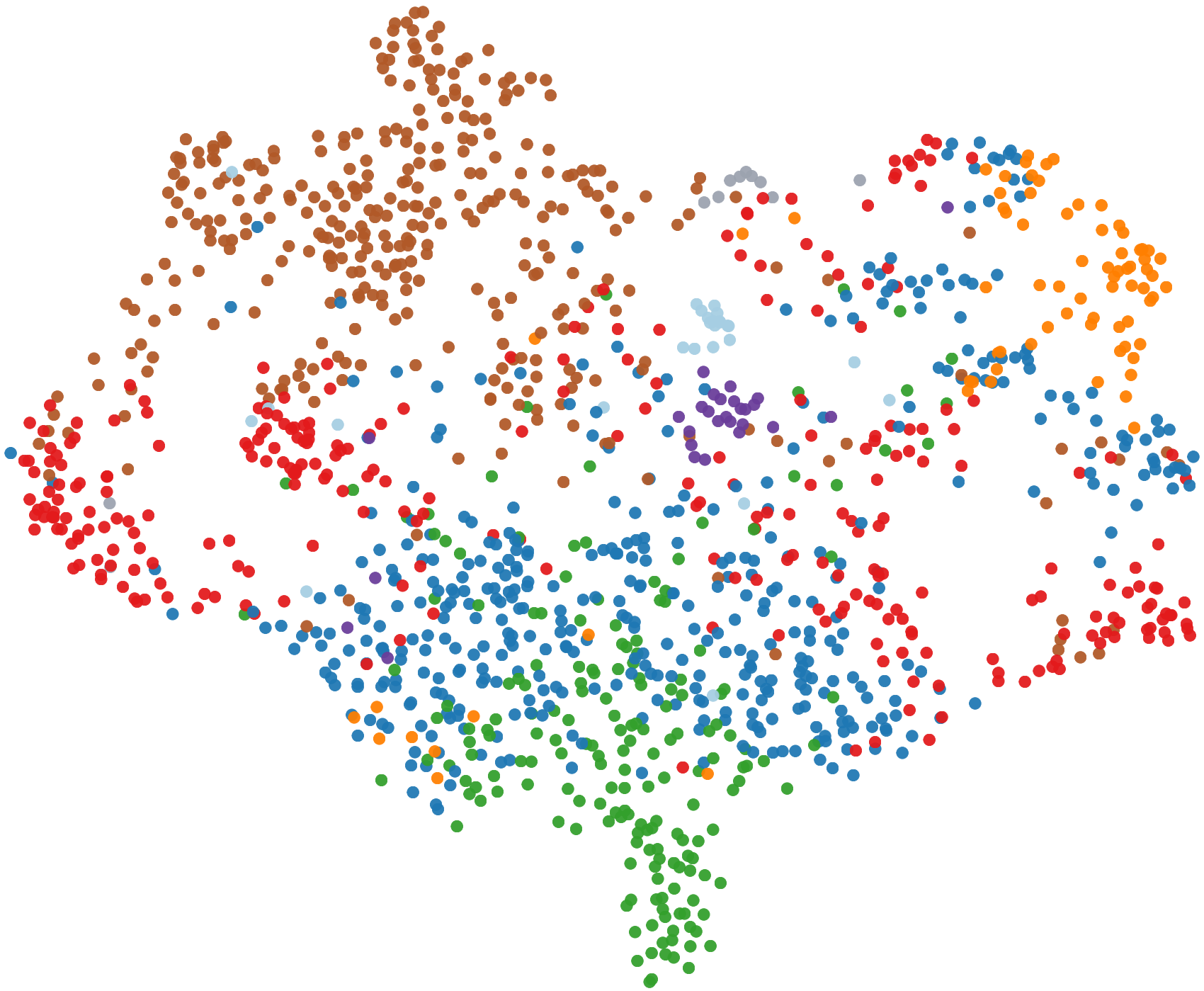
Trunk/tip/prolif fraction
0.00 0.25 0.50 0.75 1.00

Goncalves progenitor-state UMAP and full TBS tree

Terminal cells and internal tree branches shown with progenitor-state interpretation

UMAP by progenitor interpretation

Terminal cells use population-derived state classes



Goncalves adaptive-diffusion TBS tree by progenitor interpretation

Internal branches use marker/population interpretation; terminal cells use their population label

